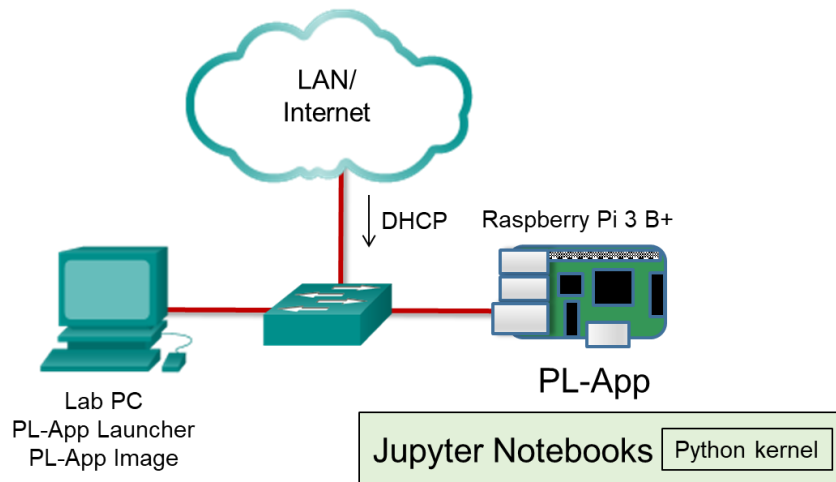


Lab - Set Up PL-App on a Raspberry Pi

Lab Topology



Objectives

- Set up a Raspberry Pi board as a The Prototyping Lab Application (PL-App) device
- Use PL-App Launcher to provision and discover PL-App devices

Background

The IoT Security lab topology uses a Raspberry Pi that is connected to a PC. The PC will be used to work with Python and Jupyter notebooks that are running on the Raspberry Pi. In addition, for some labs, the PC will run a virtual machine (VM) that interacts with the Pi.

PL-App consists of two components. The first is the PL-App launcher application that runs on the PC. It is used to provision a PL-App image that is written to a microSD card (μ SD) card for the Raspberry Pi. In addition, PL-App runs on the Raspberry Pi. It contains the Jupyter notebooks software and enables access to the notebooks and other functions of the Raspberry Pi.

With PL-App, you can access existing IoT Fundamentals labs in the Course Materials folder or write your own new applications directly on the Raspberry Pi, execute them, and monitor the output from the board with various visualizations.

Required Resources

- PL-App Launcher application
- PL-App Image file
- Wired Ethernet or Wi-Fi connection to the local-area network with DHCP
- Raspberry Pi with a power adapter
- Google Chrome or other modern web browser
- 8GB μ SD
- USB to μ SD card reader

Part 1: Setting up the PL-App

Step 1: Download and Install the PL-App Launcher.

- Navigate to www.netacad.com in your web browser and access your IoT Security class.
- From the **Student Resources** page, download the PL-App Launcher application installer for your operating system.
- After a successful download, install the PL-App Launcher application.

Note: Windows 7+ and Mac OS are supported. See the procedure at the alternate step 4 for manually installing PL-App on a Linux computer from the command line.

- After a successful installation, start the PL-App Launcher.

Step 2: Download and Install the PL-App Image.

From the Student Resources page, download the PL-App image zip file. Although the PL-App image is a file with a .zip extension, there is no need to unpack the ZIP file.

Note: The PL-App image file is large and the download speed will vary based on your Internet connection.

Step 3: Set up a micro SD card with PL-App.

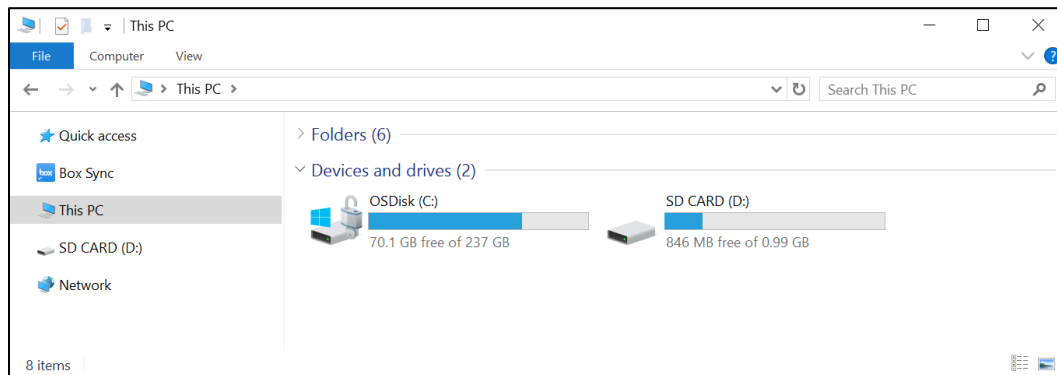
- Start the PL-App Launcher application. You should see the following screen with the **Setup a New Device** tab selected:

The screenshot shows the Cisco PL-App Launcher application window. The title bar reads "Cisco PL-App Launcher". There are two tabs: "Setup a New Device" (active) and "Available Devices". The "Setup a New Device" tab contains a numbered list of steps:

- 1** Insert the SD card reader into the USB port and select it from the dropdown menu:
[D:] [Refresh]
- 2** Select the PL-App image file that you have downloaded from NetAcad.com:
Find Image: [Browse]
- 3** Create a unique Device Name and Password for the PL-App device:
Device Name: Include your name or initials (e.g. jj-pi1984)
Device Password: Password to access PL-App on your device
- 4** Optional settings (connecting the device to an existing Wireless LAN):
WiFi SSID: _____
WiFi Password: _____
- 5** [Update Config Only] [Write Disk Image]

- Obtain a μ SD card that is at least 8 GB in size.
- Insert the μ SD card into your SD card reader and insert it into your computer.

- d. Determine the drive letter of the μ SD card using the file explorer.



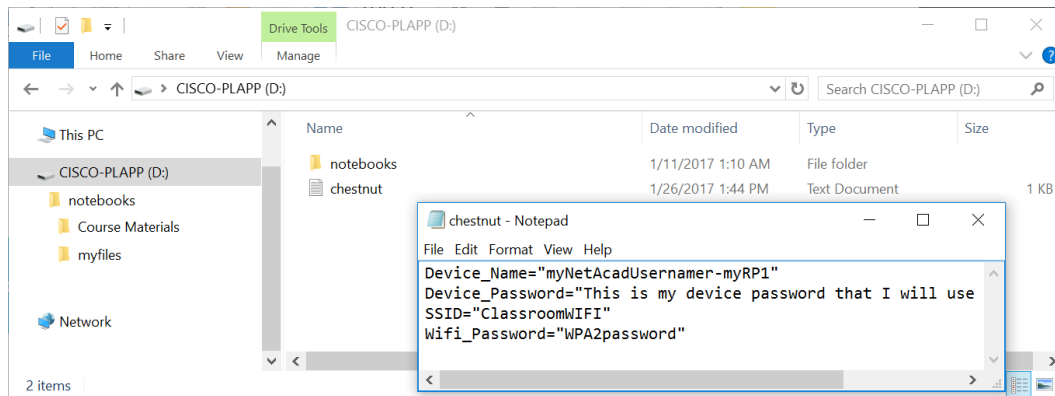
- e. In this example, it is the *D:* drive.
- f. In the PL-App Launcher, select the μ SD Card Removable Drive, in this example select the *[D:]* drive. If not present in the menu, click the **Refresh** button.
- Be careful to select the correct drive; if you select the wrong drive, you might destroy your data on the selected drive.**
- g. Click the **Find Image: Browse** button in the application and select the PL-App image zip file that you downloaded.

Step 4: Configure the Image.

- a. To personalize the PL-App installation on the μ SD card, enter the configurations section of the PL-App Launcher application and configure:
- **Device Name** - This is the name of the PL-App Raspberry Pi instance on your LAN. Make sure the selected device name is unique on the LAN, otherwise naming problems might occur. For example, you can use your name followed by a number, or your username, or initials (e.g. *myname-myRPI1*). The Device Name can contain letters (a-z), numbers (0-9) and dashes (-). The first character must be a letter.
 - **Device Password** - This is a password that you will need to enter when accessing PL-App on the Raspberry Pi. The password is stored in its clear-text form on the μ SD card. For security reasons, always use a unique password (i.e. never use the same password as your email, social media, netacad.com, etc.).
- b. If your Raspberry Pi will be connected to the network over Wi-Fi, enter the optional settings section and configure the following:
- **Wi-Fi SSID** - The name of the Wi-Fi network to join (e.g. ClassroomWiFi)
 - **Wi-Fi Password** - The WPA2 Pre-Shared Key
- c. When your settings are correct, click the **Write Disk Image** button to write the PL-App image to the selected SD card. This will overwrite all data on the card. The process will also write your configuration settings to the SD card in a file called *chestnut.txt*. Depending on the speed of your μ SD card, this process might take 5-10 minutes to complete.

Note: If the μ SD card has already been flashed with the PL-App image, and you only want to change the existing settings (e.g. Device Name, Wireless Settings, etc.), then click the **Update Config Only** button to update only the setup configuration.

- d. When the μ SD card setup is complete, use the file explorer to verify the contents of the μ SD card. You should see the μ SD card drive name has changed to CISCO-PLAPP.
- e. Explore the μ SD card drive. You should see the notebooks folder and the chestnut.txt setup file:



- o The D:\chestnut.txt file contains the setup information that was created using the PL-App Launcher. If you need to change any of these settings, you can either use the PL-App Launcher with the **Update Config Only** button to update these settings and preserve all the other contents of the SD card, or edit this file manually.
 - o The D:\notebooks\Course Materials folder contains all the *default notebook files* that were bundled with the release of the PL-App image that was flashed on the μ SD card. If you want the most recent, or unmodified, version of the lab notebooks, download the notebooks zip archive from the course page and unpack it to the D:\notebooks\Course Materials\coursename folder (the drive letter D: might be different on your computer). You can also easily backup either your own notebook files, or the Course Materials, by copying this folder to your computer.
- f. You will need secure shell (SSH) access to your Raspberry Pi in later labs. SSH can be enabled by placing an empty file named ssh in D drive (D:\ssh). When the Raspberry Pi boots, SSH is enabled when the file is found, and the file ssh is deleted from the file system.
- g. Before removing the μ SD card from your computer, make sure to safely unmount it by right-clicking the drive and selecting **Eject USB Storage**.
- h. Remove the μ SD card from your computer.

Alternate Step 4 for Advanced Linux users: Setup a μ SD card with PL-App

This is a command-line only guide for advanced Linux users. Be very careful. Incorrect use of these tools might overwrite the data on your primary hard disk. All the existing contents of the μ SD card will be overwritten in this step.

- a. Change the current working directory to the location of the PL-App image zip file that you downloaded previously:

```
cd Downloads
```
- b. Identify the device name of your μ SD card (e.g. *sdb*):

```
sudo fdisk -l
```
- c. Check if any of the μ SD card partitions are mounted using the **mount** command and then unmount them.

- d. Using **dd**, transfer the PL-App image to your μ SD card (Use caution at this step. Selecting a wrong output file "of=" parameter can overwrite your own hard disk drive and your own data):

```
unzip -p -d chestnut.img PL-App-xxxxxx.zip | sudo dd bs=1M of=/dev/{SD card device name}
```

- e. Wait for **dd** to complete. It might take 5-10 minutes without showing any output. Be patient.
- f. When completed, remove the μ SD card USB reader, wait a second, and then plug it back in to mount the newly created partition on it.
- g. In the mounted μ SD card partition (with FAT file system and 1 GB in size), create a new text file called **chestnut.txt** in the CISCO-PLAPP with the following contents:

```
Device_Name="{YOUR-NETACAD-USERNAME}-{YOUR-RPi-DEVICE-NAME}"
Device_Password="{YOUR-PL-App-PASSWORD}"
SSID="{YOUR-WIFI-SSID}"
Wifi_Password="{YOUR-WIFI-PASSWORD}"
```

Example:

```
Device_Name="myNetAcadUsername-myRpi1"
Device_Password="this is my device password"
SSID="ClassroomWIFI"
Wifi_Password="WPA2password"
```

- h. You will need secure shell (SSH) access to your Raspberry Pi in later labs. SSH can be enabled by creating an empty file named **ssh** in CISCO-PLAPP partition. When the Raspberry Pi boots, SSH is enabled when the file is found, and the file **ssh** is deleted from the file system.

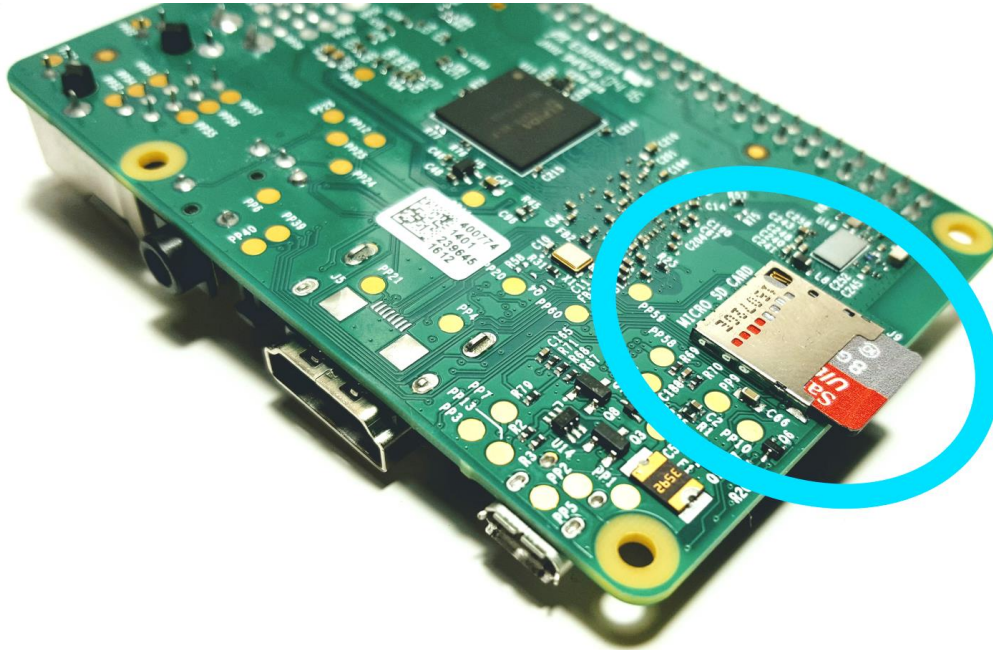
```
touch ssh
```

- i. Unmount and remove the μ SD card USB reader from your computer.
- j. Make sure your Linux computer supports mDNS discovery of hostnames:

```
sudo apt-get install libnss-mdns
```

Part 2: Accessing PL-App on a Raspberry Pi on the Network

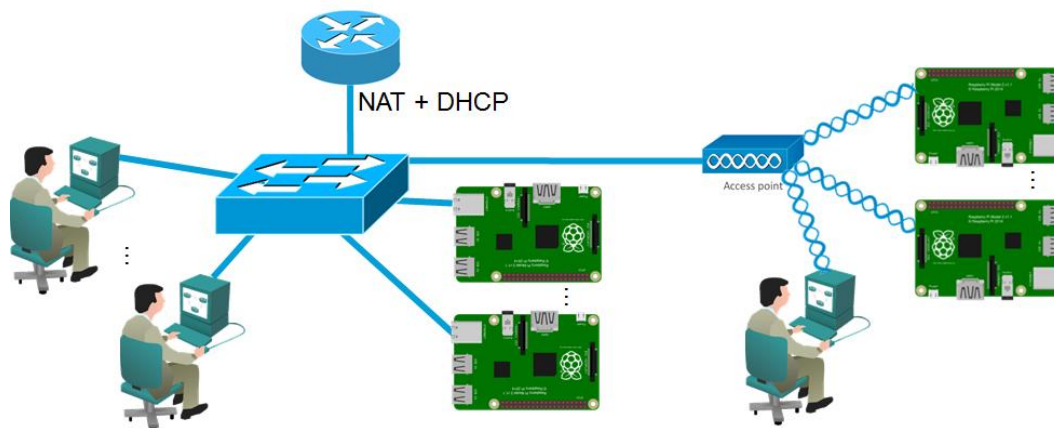
- a. Make sure that the Raspberry Pi is powered off. Insert the μ SD card into your Raspberry Pi. The μ SD card slot is on the underside of the board:



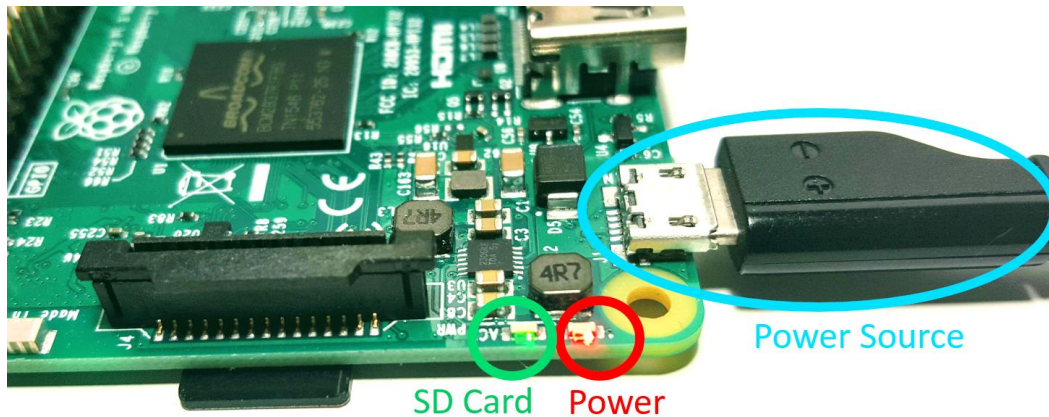
- b. Before entering or removing the μ SD card, always make sure that the Raspberry Pi is powered off.
- c. Connect the Raspberry Pi to the LAN. If you are using an Ethernet cable, connect the cable to the Raspberry Pi. Your computer and the Raspberry Pi must be in the same subnet for the PL-App Launcher discovery to work.

Note: Some labs require Internet connectivity, so a separate Internet connection is desirable.

- d. Recommended network topology for a classroom with PL-App:

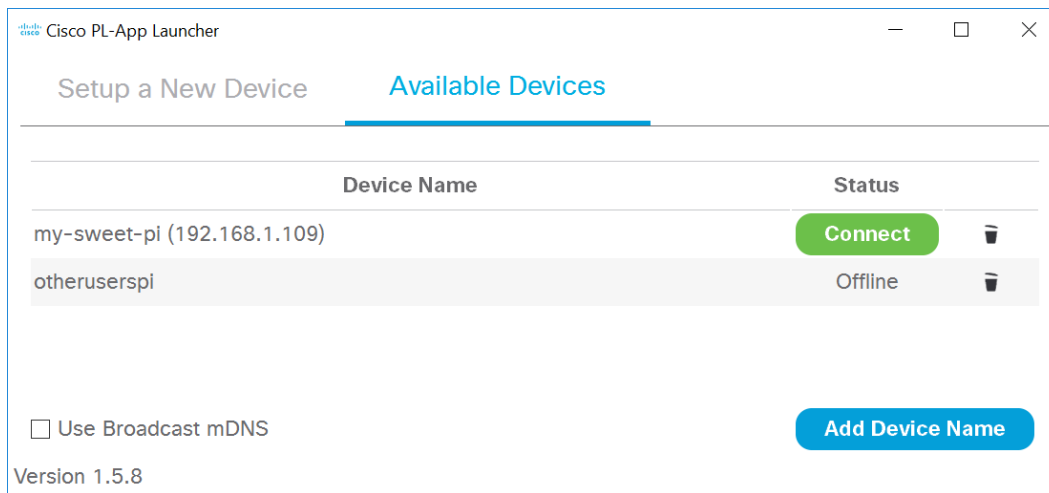


- e. Power on your Raspberry Pi by connecting the micro USB cable with a power supply that provides enough current to the Raspberry Pi board. The recommended PSU provides 2.5A to power the Raspberry Pi 3 Model B.



Note: The red Power LED must be on. If the Power LED is blinking, or is off, the power source from the PSU is not providing enough power to the board.

- f. Start the PL-App Launcher application on a PC that is on the same network as the Raspberry Pi. You should see the following screen with the **Available Devices** tab:



- g. The Available Devices tab lists all the PL-App devices that have been set up on this computer with PL-App Launcher, or that were manually added to the list using the Add Device Name button.

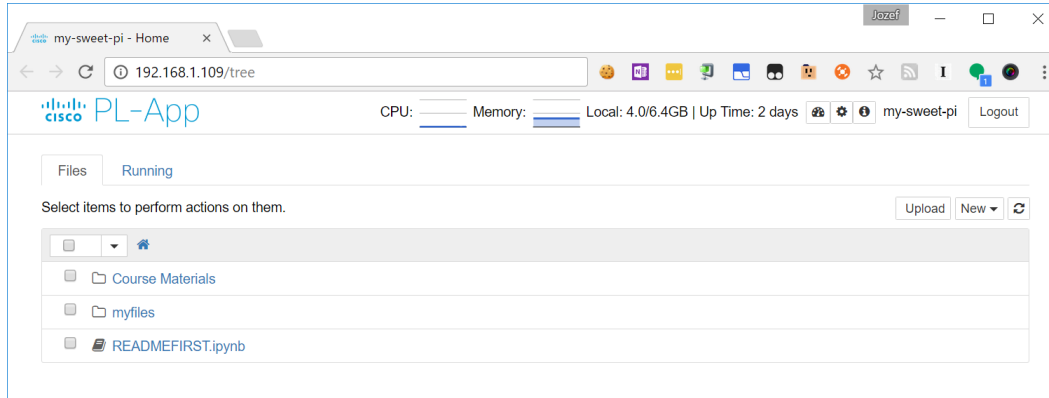
PL-App Launcher is continuously trying to discover the listed PL-App Raspberry Pi devices even if they are not connected to the network. When a device is discovered, its current IP address is shown in the list and the Connect button turns green.

Note: The first startup of a freshly written μ SD card might take longer to connect due to the initial setup process (expanding the file system of the μ SD card, etc.).

- h. Click the green **Connect** button of your device in PL-App Launcher to directly connect to the local PL-App web interface that is running on your Raspberry Pi. This opens a new web browser window and establishes a connection to the selected PL-App over HTTP.
- i. Log in to the web interface of PL-App. Use the device password you specified in PL-App Launcher in the setup process of the μ SD card.

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After a successful login, the PL-App directory browser opens the root directory of the notebooks:



This is the PL-App's representation of the notebooks folder that was created on the μ SD card.

The basic structure of the folders in the root notebooks directory is as follows:

- Course Materials: this folder contains Cisco authored labs in the form of Jupyter Notebooks.
- myfiles: this folder is your space to create your own Jupyter Notebooks and other files.

Appendix: Troubleshooting PL-App

- a. Make sure the computer with PL-App meets these criteria:
 - On the same subnet as the PL-App Raspberry Pi
 - Not connected to VPN
- b. If the onboard Red Power LED is anything but solid on (blinking, off, etc.), replace the PSU. The PSU is unable to provide enough power for the Raspberry Pi.
- c. If the onboard green μ SD card LED is constantly off or constantly on (more than 30 seconds) perform these steps:
 - 1) Check that the SD card is correctly inserted in the SD card slot on the underside of the Raspberry Pi board.
 - 2) Re-flash the μ SD card with PL-App Launcher and the PL-App image.
 - 3) Replace the μ SD card. It has been found that some SD cards are not compatible with the Raspberry Pi.
- d. Verify the μ SD card drive on a computer. Check the chestnut.txt file in the root directory of the μ SD card drive. Verify that the Device_Name specified in chestnut.txt matches the PL-App device name in the PL-App Launcher Devices tab.
- e. If connecting with Wi-Fi, perform these steps:
 - 1) Verify the network name (SSID) and corresponding WPA2 password in chestnut.txt.
 - 2) Connect to the network with an Ethernet cable (connect the computer too).
 - 3) Check if multicast is enabled on the Wi-Fi network.
- f. If connecting with an Ethernet cable, perform these steps:
 - 1) Make sure multicast is not blocked or filtered.
 - 2) Make sure DHCP is assigning IP addresses from the same subnet to the Raspberry Pi and the computer.
- g. Try to connect to a different Raspberry Pi board.